② ~~if~~ let L be a infinite Turing recognizable language.
If L is T-recognizable, there must a enumerator for it. let that enumerator be E.

~~Let's const~~

A enumerator E' exists that runs E and prints only those strings ~~in E that are~~ printed by E that are longer than, previous string printed by E'.

E' ~~consis~~ prints those strings that belong to L.
E' is an enumerator for L. ~~However~~

A TM D exists that runs E'. If any string printed by E' matches the input string then accept.
If any string printed by E' is longer than input string; reject. Since, all shorter strings have been already printed, this implies ~~...~~ ~~...~~ that input string doesn't belong to D(L).

D is a decider and it only accepts strings that belong to L.
Therefore, D(L) is a decidable subset for L.
Hence, every infinite Turing recognizable language has a Turing decidable subset.